

Multi-modal Learning Algorithms and Network Architectures for Information Extraction and Retrieval

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ABSTRACT

Large-scale (pre-)training has recently achieved great success on both uni- and multi-modal downstream evaluation tasks. However, this training paradigm generally comes with a high cost, both in the amount of compute and data needed for training. In my Ph.D. thesis, I study the problem of multi-modal learning for information extraction and retrieval, with the main focus on new learning algorithms and network architectures to make the learning process more efficient. First, I introduce a novel network architecture for bidirectional decoding for the scene text recognition (STR) task. Next, I focus on the image-caption retrieval (ICR) task. I question if the results obtained in the metric learning field generalize to the ICR task. Finally, I focus on the reduction of shortcut learning for the ICR task. I introduce latent target decoding (LTD), a novel constraint-based learning algorithm which reduces shortcut feature learning by decoding the input caption in a semantic latent space.

CCS CONCEPTS

• **Information systems** → **Image search.**

KEYWORDS

Multi-modal representation learning; Multi-modal neural networks; Contrastive learning; Multi-modal embeddings

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1 MOTIVATION

Multi-modal machine learning is a research discipline focusing on developing models that can process and relate information from multiple modalities [3]. One of the current trends in machine learning sub-fields such as computer vision [12], natural language processing [8] and multi-modal learning [30] is to apply task-agnostic pre-training in combination with large amounts of training data and big model architectures (in terms of model parameters) to obtain better performances on a down-stream evaluation task. With the

emergence of large-scale self-supervised contrastive learning, multi-modal learning and pre-training have also become more prominent [2, 14, 19, 30]. Large scale multi-modal pre-trained models show impressive results on a wide variety of downstream tasks, such as object classification [30], visual question-answering [14] and image-caption retrieval (ICR) [19].

A downside of this self-supervised learning paradigm, however, is that it requires more and more data, compute, and model parameters to improve the scores on a downstream evaluation task [1]. For example, the CLIP model [30] has been pre-trained on 400 million image-text pairs collected from the internet to train an image and text encoder. To further improve the evaluation scores on the downstream tasks, Jia et al. [19] collected a new dataset with over one billion (noisy) image alt-text pairs from the internet to train a dual-encoder architecture. This type of data collection, data processing, and pre-training is only accessible in a limited number of research environments. To keep modern machine learning research accessible to a broad and diverse group of researchers and engineers (both in industry and academia), it is important to encourage research on the understanding and development of new neural network architectures that utilize a (small) training set as efficiently as possible. Furthermore, Jia et al. [19] show that pre-training on a significantly smaller amount of clean data (i.e., 3 million curated image-text pairs) results in close-to or similar results as training on a large number (12M) of noisy scraped image-text pairs. This is an indication that scraping noisy data from the internet for pre-training does result in an improvement on the downstream evaluation task, however, it scales very slowly compared to curated data. Therefore, focusing on utilizing smaller but curated datasets more efficiently will result in bigger and faster performance gains than solely focusing on scaling-up the datasets with noisy data.

Another trend in deep learning-based machine learning is that the same network architectures are used for different data modalities and tasks. The transformer architecture [36] has been applied to, and has become the standard network architecture for, image recognition [17], language modeling [8, 15], sequence modeling [36] such as neural machine translation and many more tasks. Since the introduction of the transformer network architecture, it has become possible to easily unify multiple modalities of data into one network architecture. This opens a lot of new directions for the development and improvement of multi-modal neural methods.

In this thesis I mainly focus on two types of modalities, images and text, and two tasks, image-caption retrieval (ICR) and scene text recognition (STR). The goal of this thesis is to introduce novel multi-modal learning algorithms and network architectures. The plan is to split up this thesis into two main parts because I focus on two aspects of multi-modal learning: (1) the development of neural network architectures to combine multiple modalities of information for information extraction tasks, such as STR, and (2) the

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understanding and improvement of (low-resource) multi-modal learning algorithms, with a special focus on the image-caption retrieval (ICR) task.

2 RESULTS OBTAINED

In this section, I discuss the work and results obtained until now for my thesis. For each work, I start with the background and problem definition. Next, I introduce the novel method for the problem. Finally, I discuss the obtained results.

2.1 Bidirectional scene text recognition

Background, task, and related work. STR can be formulated as a sequence-to-sequence task. The task is to, given a sequence of embedded image regions (i.e., columns of pixels in an image), predict the sequence of characters depicted in the input image. A prominent problem in the STR task is the correct recognition of curved or rotated text in images. To mitigate the rotation invariance of convolutional neural networks, several image rectification methods have been proposed for the STR task [34, 35, 39, 40, 43]. By making use of these image rectification methods, the output predictions become more robust to perspective distortions in the input image.

Another approach to make the output sequence predictions more robust to noise in the input is bidirectional output decoding. Shi et al. [35] introduce a method for bidirectional STR. The output sequence is predicted twice by using two separate decoders: once from left-to-right (l2r) and once from right-to-left (r2l). The two decoders do not share parameters. To come to a final output prediction, the sequence with the highest decoding probability is selected as the prediction. Shi et al. [35] show that the bidirectional decoding approach outperforms a single-directed baseline model

Method. In [4], I introduce Bidirectional Scene Text Transformer (Bi-STET), a novel network architecture for bidirectional STR. Instead of having two separate decoders, one for l2r and one for r2l decoding, I unify both decoding directions into a single decoder network. I achieve this by using a transformer encoder-decoder network. Since I use the same decoder for both l2r and r2l decoding, I have to define a way to condition the output sequence on one of the two decoding directions.

To achieve this, I introduce the *direction embedding*. The direction embedding provides extra context information to the input tokens of the decoder network, similar to the position embedding in the transformer network [36]. The direction the output sequence should be decoded is represented by the direction embedding. The direction embedding is the reason I can use the same decoder for both decoding directions. The two direction embeddings are trained end-to-end with the rest of the network. Our transformer-based encoder-decoder network does not make use of any image-rectification network to handle curved and rotated text.

Results. In Table 1, I compare Bi-STET with the bidirectional STR method introduced by Shi et al. [35]. I show that our transformer-based method for 6 out of 7 evaluation sets outperforms the bidirectional STR method by Shi et al. [35]. I also show that for 6 out of 7 of the evaluation sets, Bi-STETs bidirectional decoding leads

to higher scoring sequence prediction than using a single decoding direction.

Table 1: Accuracy of left-to-right vs. right-to-left vs. bidirectional word decoding. Measured without lexicon and compared with the method by Shi et al. [35]. Dual stands for bidirectional decoding

Method		IIIT5k	SVT	IC03	IC13	IC15	SVTP	CUTE
[35], l2r		91.93	88.76	93.49	89.75	–	74.11	73.26
[35], r2l		91.43	89.96	92.79	89.95	–	73.95	74.31
[35], dual		92.27	89.5	93.60	90.54	–	74.26	74.31
Bi-STET, l2r		94.2	88.3	95.1	92.5	75.0	78.8	81.8
Bi-STET, r2l		94.1	87.9	95.3	93.4	73.2	79.5	83.6
Bi-STET, dual		94.7	89.0	96.0	93.4	75.7	80.6	82.5

In further analysis, I compare Bi-STET with state-of-the-art single directional STR methods [33, 35, 39, 39, 40, 43]. The main focus of these methods is on specific rectification network components for handling rotated or curved text in the input image. Bi-STET meets or outperforms state-of-the-art methods on 6 out of 12 evaluation experiments. This shows the strength of the transformer encoder-decoder w.r.t. to images with oriented and curved text.

2.2 Generalizability of metric learning methods to ICR

Background, task, and related work. Deep metric learning is a research field that is focusing on new loss functions to optimize encoders that map data into a shared latent space, where similar items are mapped close together in this latent space [29]. The ICR task in its nature is a contrastive discrimination task. The standard choice of optimization function for contrastive ICR methods is the triplet loss with in-batch hard negative mining [9, 13, 16, 18, 20, 21, 28] or the cross-entropy based NT-Xent loss [10, 19]. Surprisingly, not much of the progress from the metric learning field has been applied to the ICR task. In principle, the performance gain of promising metric learning functions should generalize to other contrastive learning tasks, such as ICR.

Method. In [5], I raise the question to what extent newly introduced metric learning functions generalize to the ICR task. The standard choice of optimization function for ICR methods that are trained from scratch is the triplet loss with in-batch hard negative mining [18]. In this work I examine three different loss functions for the ICR task: (i) the triplet loss [32], including in-batch hard-negative mining [18], (ii) NT-Xent loss [12], and (iii) the SmoothAP loss [7], which is a smooth approximation of the average precision metric. I optimize two baseline ICR methods, VSE++ [18] and VSRN [21] from scratch, with all three loss functions (while keeping all other training hyper-parameters the same) to examine to which extent the promising results of different metric learning functions generalize to the ICR task. For training and evaluating the model, I use two ICR benchmark datasets: MS-COCO [23] and Flickr30k [41].

In general, metric learning functions optimize the similarity between a query q and a matching (e.g., positive) candidate(s) v^+

while pushing away negative candidates $\mathbf{v}^-$ in a latent space. For each of the three metric learning functions I use in this work, I derive the gradient of that function w.r.t. $\mathbf{q}$. I observe that the gradient of all three loss functions has a similar form and that the main difference between each loss function is the number of negative candidates that are taken into account when computing the gradient. To gain a better understanding of the performance of each evaluated loss function, I introduce COCOS, which stands for *counting contributing samples*. COCOS is an analysis method to compare metric learning functions by counting how many samples contribute to the gradient w.r.t. the query representation during optimization.

Results. I examine three different metric learning functions [7, 10, 18] on the VSE++ [18] and VSRN [21] ICR methods. In contrast with findings on other contrastive or representation learning tasks, I find that the triplet loss with in-batch hard negative mining outperforms the NT-Xent [10] and SmoothAP [7] loss on both methods and both benchmark datasets. Only for the VSE++ method on the Flickr30k dataset performs SmoothAP on par with the triplet loss. Therefore, I can conclude that the triplet loss with in-batch hard negatives should still be the standard choice as a loss function for ICR methods.

To gain a better understanding of the obtained results, I apply our COCOS analysis method. I find that the best performing loss function only focuses on one (hard) negative sample when computing the gradient w.r.t. the query and therefore results in the most informative gradient. Our COCOS analysis method suggests that the underperforming loss functions take too many (non-informative) negatives into account, and therefore converge to a sub-optimal point.

2.3 Shortcut learning for multi-modal contrastive methods

Background, task, and related work. A standard ICR method makes use of two encoders: one image and one caption encoder [18, 21]. Many ICR methods are optimized by using a contrastive learning loss [9, 16, 18–21, 24, 26, 27, 37, 38, 42]. The representations learned by contrastive learning methods depend on the difficulty of the discrimination task during training [31]. The two benchmark datasets for the ICR task, MS-COCO [23] and Flickr30k [41], are relatively small in terms of training samples. This implies that it is not possible to increase the difficulty of the contrastive discrimination task during training by simply increasing the batch size. Furthermore, the smaller the batch size during training, the lower the probability of randomly sampling in-batch hard negatives. This implies that contrastive ICR methods that are trained from scratch (i.e., no pre-training on different datasets) without applying hard-negative mining strategies are prone to learn shortcut feature representations because not all information present in the training dataset is necessary to solve the contrastive optimization objective during training. One solution to increase the difficulty of the contrastive learning task during training (and therefore shortcut learning) is to apply hard negative mining over the entire training set [11]. However, due to the relatively small size of both MS-COCO and Flickr30k, this is not a feasible solution. Furthermore, hard negative mining over the entire train set can be

computationally expensive. Another possibility to avoid shortcut feature representations is to apply auto-encoding in combination with contrastive learning [22]. By reconstructing the input from the latent representation, the encoder is forced to encode as much of the input information as possible.

Method. In [6], I introduce a method to reduce shortcut learning for small-scale contrastive ICR methods. I introduce latent target decoding (LTD): a decoding framework that decodes the semantics of the input caption in a latent space. Next to the image and caption encoder, an additional caption decoder is added to the model. The purpose of this caption decoder is to reconstruct the input caption from the latent representation to prevent the image and caption encoder from learning shortcut features. Instead of decoding the caption in the input space (i.e., reconstructing the input tokens), I reconstruct the caption in a semantic latent space. The reason for decoding the caption in a latent space is that decoding the caption in the input space might not reduce shortcut features as expected. Lu et al. [25] show that strong textual decoders can reduce a reconstruction loss by mainly relying on the learned language model, which implies that the latent caption representation can still contain shortcut feature representations. Therefore, I argue that to prevent shortcut feature representations we should reconstruct the caption in a textual latent space. I achieve this by generating a *latent target* for each caption in the train set, by making use of a pre-trained general-purpose sentence encoder. The goal of the decoder is to reconstruct the input caption in the latent space of this general-purpose language encoder. I make the assumption that this sentence encoder will encode all the input information from the input caption. By reconstructing the target in a latent space, the decoder can not rely on a learned language model anymore to reconstruct the input caption, which means that the latent representation of the caption should contain as much information as possible about the target. In Figure 1, I provide an overview of the LTD method [6].

LTD results in additional loss function to the optimization objective (i.e., the reconstruction loss). Instead of implementing LTD as a dual loss, I argue to implement LTD as an optimization constraint. This implies that the main training objective is to minimize the contrastive loss, given the constraint that the reconstruction loss is below a certain threshold value.

Results. In Table 2 I compare LTD with the baseline ICR method (row 1.1/2.1), which is solely optimized by using a contrastive loss $\mathcal{L}_{con}$, on both the Flickr30k and MS-COCO benchmark dataset. I implement LTD both as a dual loss (row 1.2/2.2) and as an optimization constraint (row 1.3/2.3). Based on Table 2, I can observe that LTD optimized as a dual loss ($\mathcal{L}_{dual}$) does not convincingly outperform the baseline ICR method in terms of recall@k. In contrast, when I implement LTD as an optimization constraint, by using $\mathcal{L}_{lag}$ with a reconstruction bound of $\eta = 0.2$, I observe that LTD consistently outperforms the baseline ICR methods on both datasets with a large margin. In further analysis, I show that:

- (1) LTD generalizes to other contrastive losses;
- (2) decoding the input caption in the input space (i.e., reconstructing the input tokens instead of a latent representation) does not improve the evaluation scores;
- (3) what the role of the optimization constraint is on the learning process.

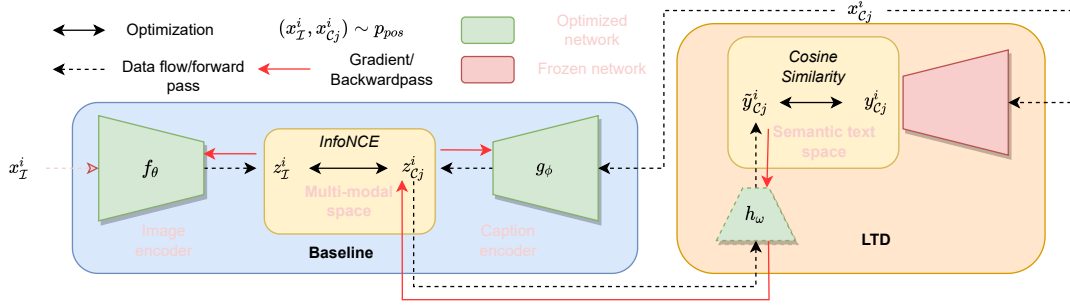


Figure 1: Overview of LTD. The baseline image-caption retrieval model consists of an image encoder and caption encoder. The encoders are trained by the contrastive InfoNCE loss. To reduce shortcut features, I add latent target decoding to the baseline ICR method. This extra decoder decodes the information of the input caption in a semantic latent space. The decoder is not used during inference, which is indicated by the dashed line around the model h_ω .

Table 2: Recall@k and r-precision evaluation scores for the Flickr30k and MS-COCO dataset. We evaluate three loss functions $\mathcal{L}_{con}$, $\mathcal{L}_{dual}$ and $\mathcal{L}_{lag}$. I use three methods, the baseline (BL) and latent target decoding (LTD).

#	Method	Loss	i2t				t2i			
			R@			R-P	R@			R-P
			1	5	10		1	5	10	
Flickr30k										
1.1	BL	$\mathcal{L}_{con}$	45.0	76.4	85.5	0.34	33.2	65.5	76.3	-
1.2	LTD	$\mathcal{L}_{dual}, \beta=1$	45.8	75.3	84.6	0.34	33.7	66.0	77.0	-
1.3	LTD	$\mathcal{L}_{lag}, \eta=0.2$	49.7	78.9	86.7	0.37	37.5	68.6	79.5	-
MS-COCO										
2.1	BL	$\mathcal{L}_{con}$	32.9	64.3	76.5	0.24	23.8	53.3	66.9	-
2.2	LTD	$\mathcal{L}_{dual}, \beta=1$	34.4	64.1	76.7	0.25	24.5	54.0	67.5	-
2.3	LTD	$\mathcal{L}_{lag}, \eta=0.2$	35.1	66.7	78.6	0.26	25.6	55.0	68.7	-

3 ON-GOING AND FUTURE WORK

In this section I will discuss on-going and future work planned for this thesis.

Problem definition. The final work of this thesis will be a follow-up work to the LTD I introduce in [6]. One of the open questions I discuss in [6] is how to measure the reduction of shortcut features properly. In [6], the reduction of shortcut features is measured by the increase in evaluation performances on the downstream evaluation task (i.e., retrieval), which is mainly an indication of the generalizability of the learned embeddings. To get a better understanding of how LTD is reducing shortcut features, I introduce artificial shortcuts in the training data.

Proposed method. The shortcuts I introduce in this work are artificial, which implies that I am able to manipulate these features. Therefore, I can measure the influence of these features on the downstream evaluation task directly. The goal is to make these shortcuts extremely clear and prominent during training, with the result that the caption and image encoder will mainly rely on artificial shortcuts to solve the contrastive training objective. I

train two ICR models with artificial shortcuts: one baseline that is optimized only with a contrastive loss and one contrastive model in combination with LTD.

Target results. I hypothesize that when evaluating the contrastive baseline without artificial shortcuts, the evaluation performance will drop significantly due to the fact that the artificial shortcuts are not present anymore during the evaluation. However, for the model trained in combination with LTD, I do not expect to see this performance drop due to the effect of LTD. If I do not observe the drop in evaluation performance for the ICR model that has been trained with LTD, I can conclude that this model does not solely rely on the shortcuts present in the training data, but also takes additional information into account to rank candidates.

4 CONCLUSION

In this thesis, I focus on novel learning algorithms and network architectures for multi-modal retrieval and information extraction. So far, I have introduced a novel transformer-based network architecture for bidirectional STR. For the ICR task, I first look into the question if lessons for the metric learning field generalize the ICR task. Next, I propose LTD, a new method for the ICR task to reduce the learning of shortcut feature representations in a low-resource setting. In the near future, I intend to look into the problem of artificial shortcut features for ICR. This with the goal to directly measure the effect of LTD on the reduction of shortcut features. To facilitate the reproducibility of our work [4–6] me and my co-authors have released the code for each paper.¹

I hope that this work can inspire new research on (1) efficient contrastive multi-modal learning algorithms, and (2) novel network architectures that efficiently can combine multiple modalities of information without relying on a vast amount of training data scraped from the internet.

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¹<https://github.com/MauritsBleeler/BiStet>, ecir-2022-reproducibility-bleeker, keep-the-caption-info

REFERENCES

- [1] Samira Abnar, Mostafa Dehghani, Behnam Neyshabur, and Hanie Sedghi. 2021. Exploring the limits of large scale pre-training. *arXiv preprint arXiv:2110.02095* (2021).
- [2] Jean-Baptiste Alayrac, Jeff Donahue, Pauline Luc, Antoine Miech, Iain Barr, Yana Hasson, Karel Lenc, Arthur Mensch, Katie Millican, Malcolm Reynolds, et al. 2022. Flamingo: a Visual Language Model for Few-Shot Learning. *arXiv preprint arXiv:2204.14198* (2022).
- [3] Tadas Baltrušaitis, Chaitanya Ahuja, and Louis-Philippe Morency. 2018. Multi-modal machine learning: A survey and taxonomy. *IEEE transactions on pattern analysis and machine intelligence* 41, 2 (2018), 423–443.
- [4] Maurits Bleeker and Maarten de Rijke. 2020. Bidirectional Scene Text Recognition with a Single Decoder. In *ECAI 2020: 24th European Conference on Artificial Intelligence*. IOS Press, 2664–2672.
- [5] Maurits Bleeker and Maarten de Rijke. 2022. Do Lessons from Metric Learning Generalize to Image-Caption Retrieval? In *ECIR 2022: 44th European Conference on Information Retrieval*. Springer.
- [6] Maurits Bleeker, Andrew Yates, and Maarten de Rijke. 2022. Keep the Caption Information: Preventing Shortcut Learning in Contrastive Image-Caption Retrieval. *arXiv preprint arXiv:2204.13382* (2022).
- [7] Andrew Brown, Weidi Xie, Vicky Kalogeiton, and Andrew Zisserman. 2020. Smooth-AP: Smoothing the path towards large-scale image retrieval. In *European Conference on Computer Vision (ECCV)*. Springer, 677–694.
- [8] Tom Brown, Benjamin Mann, Nick Ryder, Melanie Subbiah, Jared D Kaplan, Prafulla Dhariwal, Arvind Neelakantan, Pranav Shyam, Girish Sastry, Amanda Askell, et al. 2020. Language models are few-shot learners. *Advances in neural information processing systems* 33 (2020), 1877–1901.
- [9] Hui Chen, Guiguang Ding, Xudong Liu, Zijia Lin, Ji Liu, and Jungong Han. 2020. IMRAM: Iterative Matching With Recurrent Attention Memory for Cross-Modal Image-Text Retrieval. In *2020 IEEE/CVF Conference on Computer Vision and Pattern Recognition, CVPR 2020, Seattle, WA, USA, June 13-19, 2020*. Computer Vision Foundation / IEEE, 12652–12660. <https://doi.org/10.1109/CVPR42600.2020.01267>
- [10] Tianlang Chen, Jiajun Deng, and Jiebo Luo. 2020. Adaptive Offline Quintuplet Loss for Image-Text Matching. In *European Conference on Computer Vision (ECCV)*. Springer, 549–565.
- [11] Tianlang Chen, Jiajun Deng, and Jiebo Luo. 2020. Adaptive Offline Quintuplet Loss for Image-Text Matching. In *Computer Vision - ECCV 2020 - 16th European Conference, Glasgow, UK, August 23-28, 2020, Proceedings, Part XIII (Lecture Notes in Computer Science, Vol. 12358)*, Andrea Vedaldi, Horst Bischof, Thomas Brox, and Jan-Michael Frahm (Eds.). Springer, 549–565. https://doi.org/10.1007/978-3-030-58601-0_33
- [12] Ting Chen, Simon Kornblith, Mohammad Norouzi, and Geoffrey Hinton. 2020. A simple framework for contrastive learning of visual representations. In *International conference on machine learning*. PMLR, 1597–1607.
- [13] Ting Chen and Lala Li. 2020. Intriguing Properties of Contrastive Losses. *arXiv preprint arXiv:2011.02803* (2020).
- [14] Yen-Chun Chen, Linjie Li, Licheng Yu, Ahmed El Kholy, Faisal Ahmed, Zhe Gan, Yu Cheng, and Jingjing Liu. 2020. Uniter: Universal image-text representation learning. In *European conference on computer vision*. Springer, 104–120.
- [15] Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. 2018. Bert: Pre-training of deep bidirectional transformers for language understanding. *arXiv preprint arXiv:1810.04805* (2018).
- [16] Haiwen Diao, Ying Zhang, Lin Ma, and Huchuan Lu. 2021. Similarity Reasoning and Filtration for Image-Text Matching. In *Thirty-Fifth AAAI Conference on Artificial Intelligence, AAAI 2021, Thirty-Third Conference on Innovative Applications of Artificial Intelligence, IAAI 2021, The Eleventh Symposium on Educational Advances in Artificial Intelligence, EAAI 2021, Virtual Event, February 2-9, 2021*. AAAI Press, 1218–1226. <https://ojs.aaai.org/index.php/AAAI/article/view/16209>
- [17] Alexey Dosovitskiy, Lucas Beyer, Alexander Kolesnikov, Dirk Weissenborn, Xi-aohua Zhai, Thomas Unterthiner, Mostafa Dehghani, Matthias Minderer, Georg Heigold, Sylvain Gelly, et al. 2020. An image is worth 16x16 words: Transformers for image recognition at scale. *arXiv preprint arXiv:2010.11929* (2020).
- [18] Fartash Faghri, David J Fleet, Jamie Ryan Kiros, and Sanja Fidler. 2018. VSE++: Improving Visual-Semantic Embeddings with Hard Negatives. In *Proceedings of the British Machine Vision Conference (BMVC)*.
- [19] Chao Jia, Yinfei Yang, Ye Xia, Yi-Ting Chen, Zarana Parekh, Hieu Pham, Quoc V Le, Yunhsuan Sung, Zhen Li, and Tom Duerig. 2021. Scaling up visual and vision-language representation learning with noisy text supervision. *arXiv preprint arXiv:2102.05918* (2021).
- [20] Kuang-Huei Lee, Xi Chen, Gang Hua, Houdong Hu, and Xiaodong He. 2018. Stacked Cross Attention for Image-Text Matching. In *Computer Vision - ECCV 2018 - 15th European Conference, Munich, Germany, September 8-14, 2018, Proceedings, Part IV (Lecture Notes in Computer Science, Vol. 11208)*, Vittorio Ferrari, Martial Hebert, Cristian Sminchisescu, and Yair Weiss (Eds.). Springer, 212–228. https://doi.org/10.1007/978-3-030-01225-0_13
- [21] Kunpeng Li, Yulun Zhang, Kai Li, Yuanyuan Li, and Yun Fu. 2019. Visual semantic reasoning for image-text matching. In *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*. 4654–4662.
- [22] Tianhong Li, Lijie Fan, Yuan Yuan, Hao He, Yonglong Tian, Rogerio Feris, Piotr Indyk, and Dina Katabi. 2020. Making contrastive learning robust to shortcuts. *arXiv preprint arXiv:2012.09962* (2020).
- [23] Tsung-Yi Lin, Michael Maire, Serge Belongie, James Hays, Pietro Perona, Deva Ramanan, Piotr Dollár, and C Lawrence Zitnick. 2014. Microsoft COCO: Common objects in context. In *European Conference on Computer Vision*. Springer, 740–755.
- [24] Chunxiao Liu, Zhendong Mao, Tianzhu Zhang, Hongtao Xie, Bin Wang, and Yongdong Zhang. 2020. Graph Structured Network for Image-Text Matching. In *2020 IEEE/CVF Conference on Computer Vision and Pattern Recognition, CVPR 2020, Seattle, WA, USA, June 13-19, 2020*. Computer Vision Foundation / IEEE, 10918–10927. <https://doi.org/10.1109/CVPR42600.2020.01093>
- [25] Shuqi Lu, Chenyan Xiong, Di He, Guolin Ke, Waleed Malik, Zhicheng Dou, Paul Bennett, Tie-Yan Liu, and Arnold Overwijk. 2021. Less is More: Pre-training a Strong Siamese Encoder Using a Weak Decoder. *CoRR abs/2102.09206* (2021). <https://arxiv.org/abs/2102.09206>
- [26] Nicola Messina, Giuseppe Amato, Andrea Esuli, Fabrizio Falchi, Claudio Gennaro, and Stéphane Marchand-Maillet. 2020. Fine-grained Visual Textual Alignment for Cross-Modal Retrieval using Transformer Encoders. *CoRR abs/2008.05231* (2020). <https://arxiv.org/abs/2008.05231>
- [27] Nicola Messina, Fabrizio Falchi, Andrea Esuli, and Giuseppe Amato. 2020. Transformer Reasoning Network for Image-Text Matching and Retrieval. In *25th International Conference on Pattern Recognition, ICPR 2020, Virtual Event / Milan, Italy, January 10-15, 2021*. IEEE, 5222–5229. <https://doi.org/10.1109/ICPR48806.2021.9413172>
- [28] Nicola Messina, Fabrizio Falchi, Andrea Esuli, and Giuseppe Amato. 2021. Transformer reasoning network for image-text matching and retrieval. In *International Conference on Pattern Recognition (ICPR)*. IEEE, 5222–5229.
- [29] Kevin Musgrave, Serge Belongie, and Ser-Nam Lim. 2020. A metric learning reality check. In *European Conference on Computer Vision*. Springer, 681–699.
- [30] Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, et al. 2021. Learning transferable visual models from natural language supervision. In *International Conference on Machine Learning*. PMLR, 8748–8763.
- [31] Joshua Robinson, Li Sun, Ke Yu, Kayhan Batmanghelich, Stefanie Jegelka, and Suvrit Sra. 2021. Can contrastive learning avoid shortcut solutions? *arXiv preprint arXiv:2106.11230* (2021).
- [32] Florian Schroff, Dmitry Kalenichenko, and James Philbin. 2015. Facenet: A unified embedding for face recognition and clustering. In *Proceedings of the IEEE conference on computer vision and pattern recognition*. 815–823.
- [33] Fenfen Sheng, Zheneng Chen, and Bo Xu. 2018. NRTR: A No-Recurrence Sequence-to-Sequence Model For Scene Text Recognition. *arXiv preprint arXiv:1806.00926* (2018).
- [34] Baoguang Shi et al. 2016. Robust scene text recognition with automatic rectification. In *CVPR*. 4168–4176.
- [35] Baoguang Shi et al. 2018. Aster: An attentional scene text recognizer with flexible rectification. *IEEE transactions on pattern analysis and machine intelligence* (2018).
- [36] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N Gomez, Łukasz Kaiser, and Illia Polosukhin. 2017. Attention is all you need. *Advances in neural information processing systems* 30 (2017).
- [37] Sijin Wang, Ruipeng Wang, Ziwei Yao, Shiguang Shan, and Xilin Chen. 2020. Cross-modal Scene Graph Matching for Relationship-aware Image-Text Retrieval. In *IEEE Winter Conference on Applications of Computer Vision, WACV 2020, Snowmass Village, CO, USA, March 1-5, 2020*. IEEE, 1497–1506. <https://doi.org/10.1109/WACV45572.2020.9093614>
- [38] Zihao Wang, Xihui Liu, Hongsheng Li, Lu Sheng, Junjie Yan, Xiaogang Wang, and Jing Shao. 2019. CAMP: Cross-Modal Adaptive Message Passing for Text-Image Retrieval. In *2019 IEEE/CVF International Conference on Computer Vision, ICCV 2019, Seoul, Korea (South), October 27 - November 2, 2019*. IEEE, 5763–5772. <https://doi.org/10.1109/ICCV.2019.00586>
- [39] MingKun Yang, Yushuo Guan, Minghui Liao, Xin He, Kaigui Bian, Song Bai, Cong Yao, and Xiang Bai. 2019. Symmetry-constrained Rectification Network for Scene Text Recognition. *arXiv preprint arXiv:1908.01957* (2019).
- [40] Xiao Yang, Dafang He, Zihan Zhou, Daniel Kifer, and C Lee Giles. 2017. Learning to Read Irregular Text with Attention Mechanisms. In *IJCAI*. 3280–3286.
- [41] Peter Young, Alice Lai, Micah Hodosh, and Julia Hockenmaier. 2014. From image descriptions to visual denotations: New similarity metrics for semantic inference over event descriptions. *Transactions of the Association for Computational Linguistics* 2 (2014), 67–78.
- [42] Tan Yu, Yi Yang, Yi Li, Lin Liu, Hongliang Fei, and Ping Li. 2021. Heterogeneous Attention Network for Effective and Efficient Cross-modal Retrieval. In *SIGIR '21: The 44th International ACM SIGIR Conference on Research and Development in Information Retrieval, Virtual Event, Canada, July 11-15, 2021*, Fernando Diaz, Chirag Shah, Torsten Suel, Pablo Castells, Rosie Jones, and Tetsuya Sakai (Eds.). ACM, 1146–1156. <https://doi.org/10.1145/3404835.3462924>
- [43] Fangneng Zhan and Shijian Lu. 2018. Esir: End-to-end scene text recognition via iterative image rectification. *arXiv preprint arXiv:1812.05824* (2018).